FAST NUCES, Islamabad Campus

Granite

AI-agent
investigation workbench

Find why AI agents fail.

Registered: A Self-Healing Multi-Agent System
for Semantic Error Diagnosis, Verified Repair and Recovery

PROJECT GOALS

- Detect silent failures
- Locate their source
- Verify proposed fixes

For engineers & QA teams
People stay in control

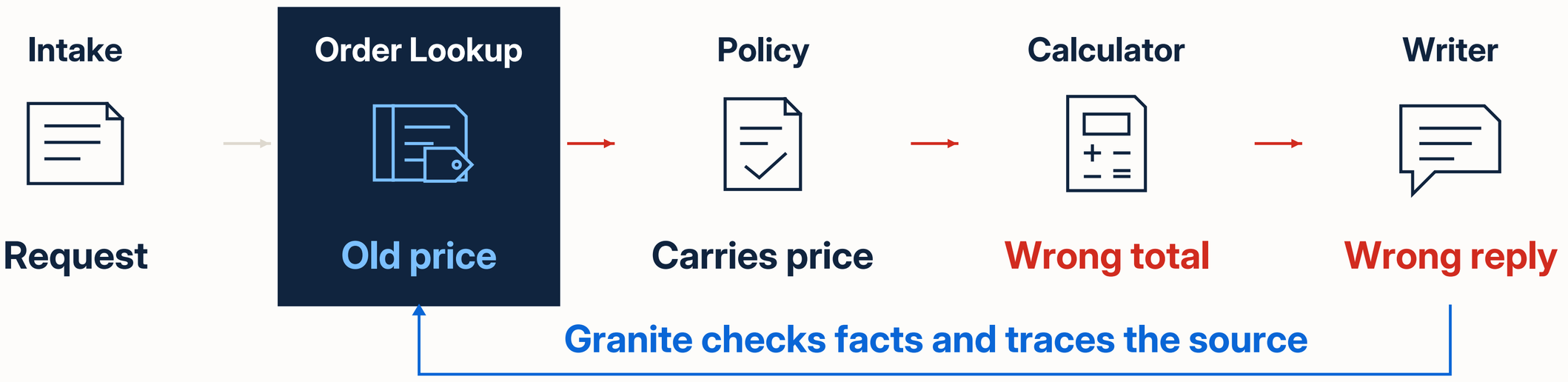
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F26-060 · BS (DS) · FYP 2026

01 / THE SILENT FAILURE

One bad value. A wrong AI answer.

The workflow finishes without raising an error.



Refund Desk · stale-price fault · simplified illustration

02 / CONTROLLED REPLAY · IMPLEMENTED

Prove the cause. Test the correction.

Source corrected	✓ ✓ ✓ ✓ ✓	5/5 correct
Source unchanged	✗ ✗	0/2 correct

Cause verified
Human review
Evidence saved

Recorded case · five correction replays, two controls.

03 / EARLY FEASIBILITY · 7 OCT 2026

10/10 faulty runs detected

9/10 origins verified

10 tasks · one fault · one small model · four tasks not blind.

04 / VISION & TENTATIVE FYP-1 PLAN

The finished system: evidence to action.

Goal: faster diagnosis and reusable tests for QA.

BUILT
1 · Investigate
Refund Desk workbench
Checks · trace · replay

PLANNED
2 · Guide repairs
One-click investigation
Check-derived fixes

PLANNED
3 · Extend
Retail Agent · imports
Regression-test export

TOOLS & TECHNOLOGIES

Python · FastAPI · React · TypeScript · Ollama